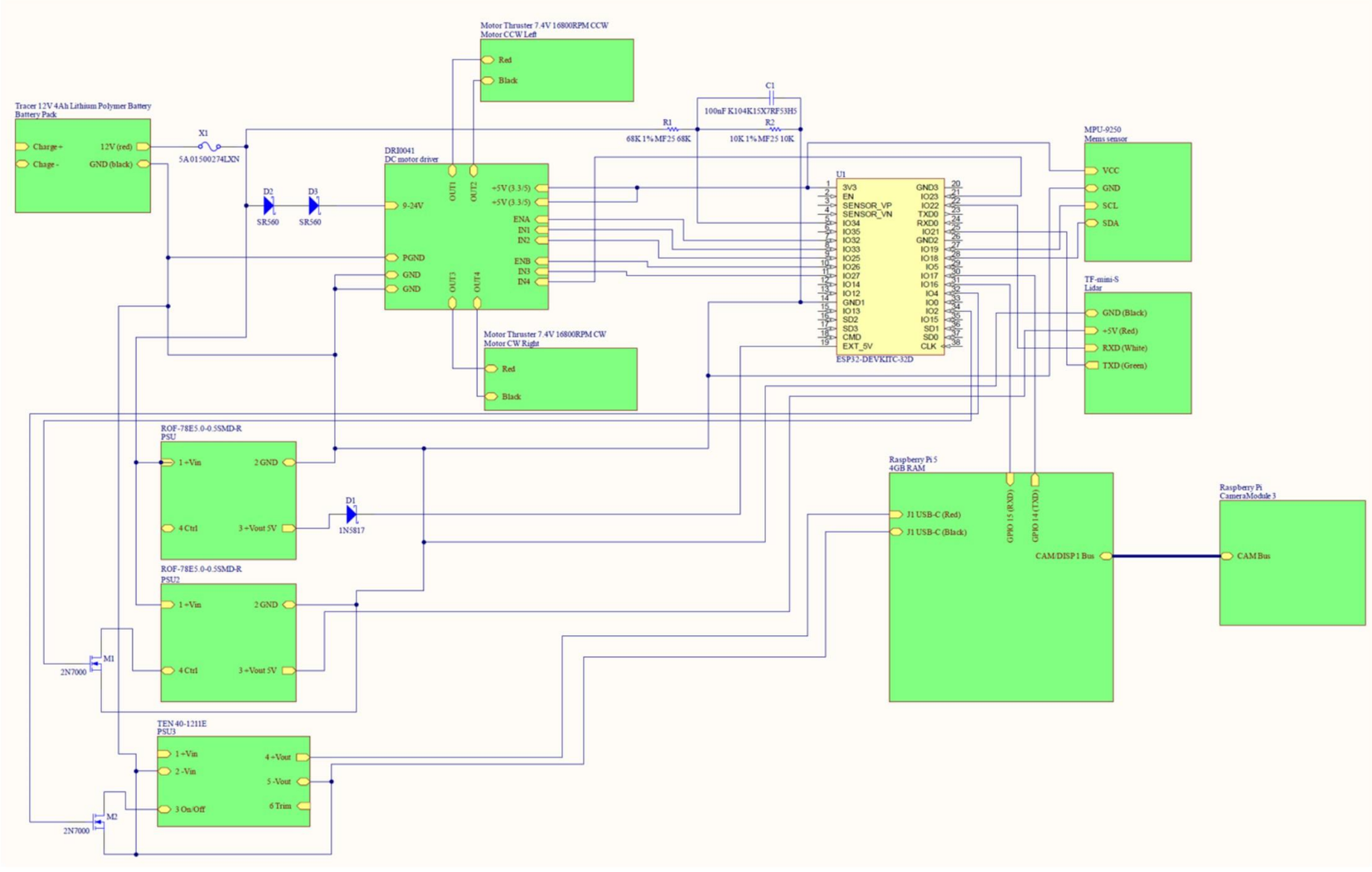
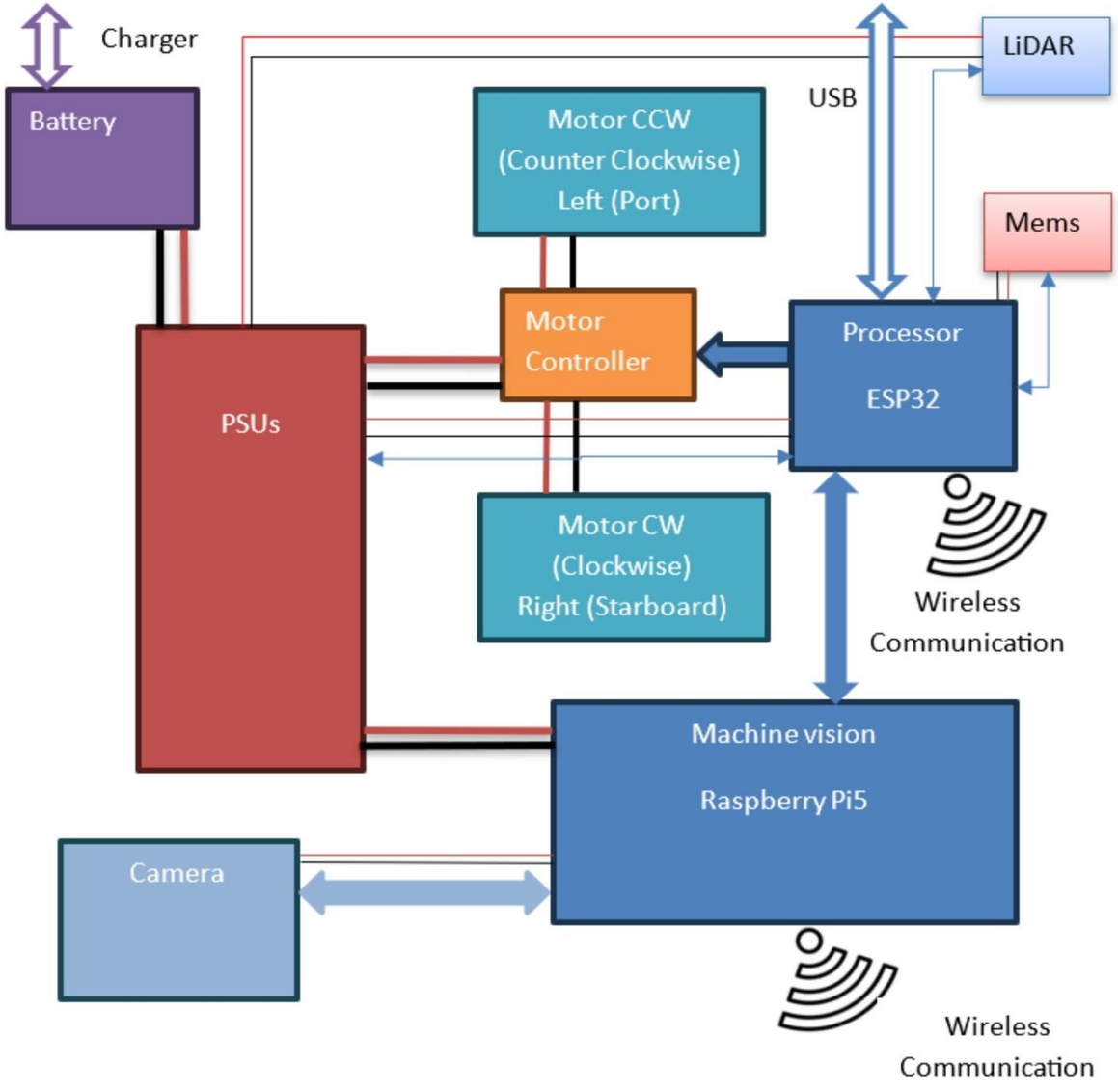
Electronics Design -
I designed the circuit, system architecture and internal layout. A 12 V LiPo battery feeds three power-supply modules; an ESP32 drives the two thrusters through a motor driver and reads an IMU and a LiDAR; a Raspberry Pi 5 with a camera handles machine vision. The wiring-loom plan maps every cable route inside the hull.

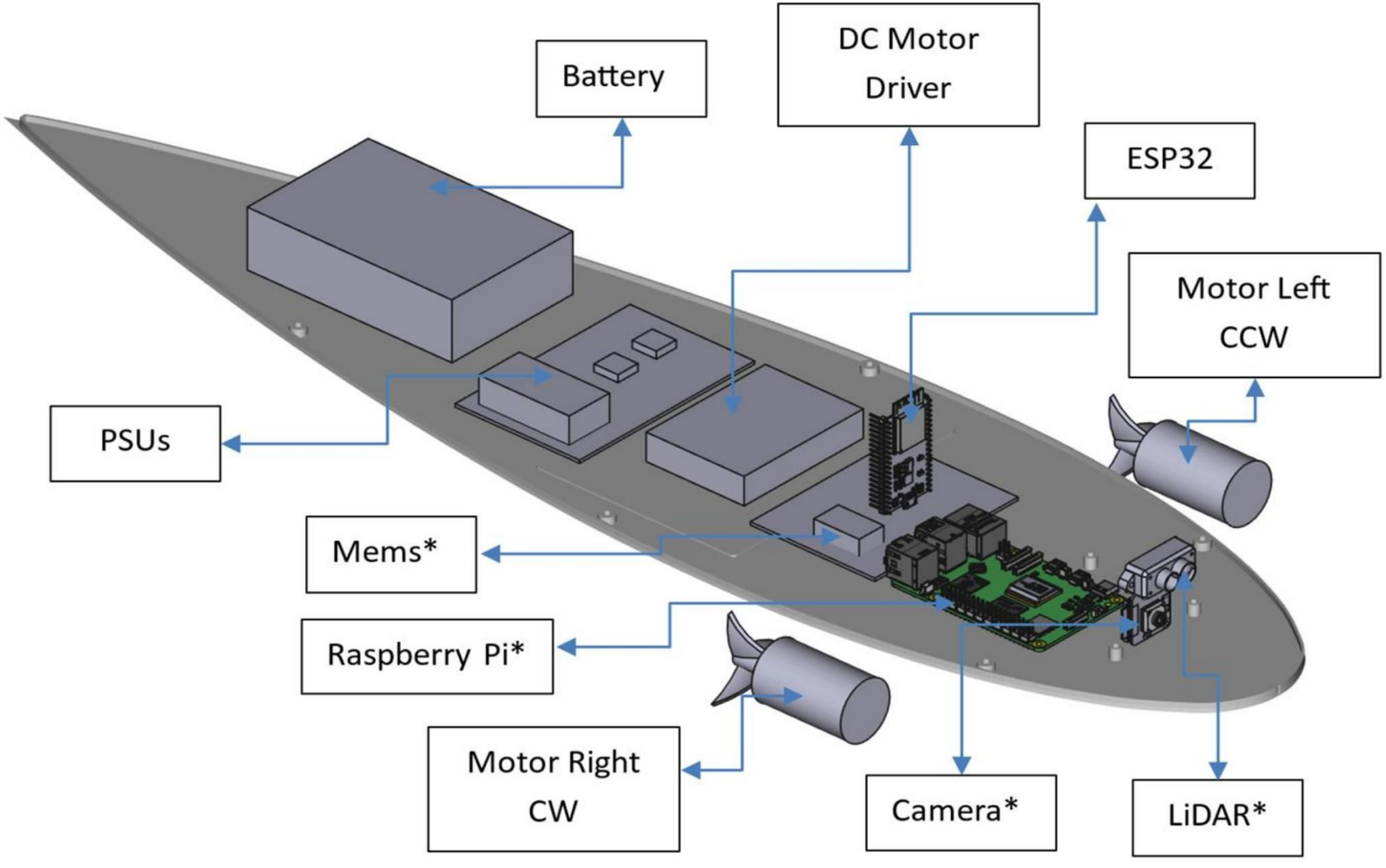
Marine Collection Vessel - Robot Schematic



Marine Collection Vessel - Robot Block Diagram



Marine Collection Vessel - Robot Components



Marine Collection Vessel - Robot Wiring Loom

